

Assignment 1: Empathy – Understanding User Needs

Updated Feature Included: Senior & Alumni Interaction Module

Project Topic: PCCOER Student Support Portal

A. Target User Identification

User Group:

- Students
- Faculty Members
- College Administration Staff
- Librarian

Age Range:

- 18–25 years (Students)
- 25–55 years (Faculty/Admin)

Context of Use:

Users interact with the portal to register complaints, submit suggestions, access mentor/counselor support, explore campus facilities, check library book availability and view college events or hackathons. The system is mainly used during college hours or whenever issues arise.

Environment (Physical/Digital):

- Physical: College campus, classrooms, hostel, library, gym, sports area
 - Digital: Web portal accessible through desktop and mobile devices
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B. User Role and Responsibility

Primary Tasks Performed by the User:

- Register complaints
- Upload images/videos as proof
- Track complaint status
- Submit suggestions
- View events and hackathons
- Request mentor/counselor support
- Explore facilities and library books

Frequency of Use:

- Daily / Occasional
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C. Key Challenges Faced by the User

Major Difficulties or Pain Points:

- Delayed complaint resolution
 - Lack of transparency in complaint tracking
 - Difficulty contacting concerned authorities
 - No centralized student support system
 - Lack of awareness about events and campus facilities
 - Difficulty checking library book availability
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D. Notes and Observations

- Students prefer online complaint registration instead of manual forms.
 - Students want quick updates regarding complaints.
 - Many students are unaware of ongoing hackathons and events.
 - Students need one centralized platform for complaints, support and campus services.
 - Students also want interaction with seniors and alumni for placement guidance and mentorship.
 - Faculty members prefer organized complaint handling.
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E. Empathy Methods Used

✓ Interview

Description:

Interviews were conducted with:

- 6 students
- 2 faculty members
- 1 administrative staff member

Questions asked:

- What issues do students commonly face?
 - How are complaints currently handled?
 - What problems exist in the current system?
 - What features should a student support portal include?
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✓ Survey

Description:

- Number of respondents: 30 students
- Questions included:
 - Satisfaction with current complaint process
 - Awareness of college events
 - Need for mentor/counselor support
 - Preferred complaint submission method

Purpose: To identify student expectations and common campus issues.

✓ Observation

Description:

Observation was conducted in:

- College office
- Library
- Hostel
- Event notice board areas

Observed behavior:

- Students repeatedly visiting offices for updates
 - Manual records causing delays
 - Students missing event information
 - Difficulty finding library book availability information
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Key Insights Gained from These Methods

1. Students want a centralized digital support system.
 2. Users need real-time complaint tracking.
 3. Students prefer quick access to events, facilities and library information.
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F. Key User Quotes / Observations

Sr No.	Method Used	User Quote / Observation	Insight
1	Interview	"We don't know whether our complaint is resolved or not."	Need for status tracking
2	Interview	"Sometimes complaints are ignored."	Need for accountability

Sr No.	Method Used	User Quote / Observation	Insight
3	Survey	Most students prefer online systems	Digital portal preferred
4	Observation	Students repeatedly visit offices for updates	Communication gap exists
5	Interview	"We often miss hackathon updates."	Need for event section
6	Survey	Students want mentor/counselor support	Student support needed
7	Observation	Manual records are difficult to manage	Need for automation
8	Interview	"Book availability should be visible online."	Need for library module

G. Pain Points Identified

Pain Point 1:

Delayed complaint resolution due to manual processes.

Pain Point 2:

No proper tracking system for complaint status.

Pain Point 3:

Lack of centralized student support and communication.

Pain Point 4:

Students are unaware of campus events and facility information.

H. Empathy Map

Says

- "I want quick complaint updates."
- "There should be one platform for all student services."
- "We miss important event announcements."

Thinks

- "Will my complaint be taken seriously?"
- "Why is the process so slow?"

- “College services should be more accessible online.”

Does

- Visits office repeatedly
- Calls faculty/admin for updates
- Checks multiple sources for events and support

Feels

- Frustrated
 - Ignored
 - Unsatisfied
 - Hopeful for better digital support
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I. Evidence

Attach:

- Screenshots of surveys
- Interview photographs
- Sample complaint forms
- UI mockups/screenshots

Assignment 2: Define Problem

Updated Feature Included: Senior & Alumni Interaction Module

Project Topic: PCCOER Student Support Portal

A. Problem Background

What problem exists?

Students face difficulties while reporting complaints, accessing support services, finding event information and checking campus resources because the current process is manual, scattered and lacks transparency. There is no centralized platform for student support, communication and mentorship interaction with seniors and alumni.

B. Who Faces It?

The primary users affected are:

- Students
 - Faculty members
 - College administration staff
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C. Where and When Does It Occur?

The problem occurs:

- Inside college campuses
 - In classrooms, hostels, library, sports facilities and event areas
 - During academic hours or whenever students face issues or require support
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D. Problem Statement

College students and faculty members face difficulties in reporting complaints, accessing student support services and obtaining campus-related information because the current system is manual, unorganized and lacks transparency. This leads to delayed resolutions, poor communication, lack of awareness about events and inconvenience in accessing campus services. As a result, students experience frustration and reduced satisfaction with campus support systems. An ideal solution should provide a centralized digital platform that improves complaint management, communication, support accessibility and campus information services.

How Might We (HMW) Question

How might we help students and faculty members access support services, register complaints and explore campus resources efficiently through one centralized platform?

E. Constraints

Technical Constraints

- Frontend and backend integration complexity
 - Database management requirements
 - File upload handling for images/videos
 - Internet connectivity limitations
 - Data security and authentication management
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Financial Constraints

- Limited project development budget
 - Server and maintenance costs
 - Resource limitations
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Time Constraints

- Limited academic project duration
 - Need for quick development and testing
 - Timely complaint response expectations
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User-Related Constraints

- Different technical skill levels among users
 - Resistance to switching from manual systems
 - Need for simple and user-friendly interface
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F. Success Criteria

How Will You Know the Problem Is Solved?

The problem will be considered solved if:

- Complaints can be registered online easily
 - Users can track complaint status in real time
 - Students can access support services efficiently
 - Events and campus information become easily accessible
 - Library book availability can be checked online
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User Experience Improvement

- Faster complaint submission and tracking
 - Improved communication with administration
 - Easy access to events, facilities and support services
 - Better student engagement
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Reduction in Pain Points

- Reduced delays in complaint handling
- Elimination of repeated office visits
- Improved transparency and accountability

- Better access to student support resources

Efficiency or Performance Gains

- Faster complaint processing
 - Reduced paperwork
 - Centralized management of student services
 - Better organization of campus information
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User Satisfaction and Acceptance

- Students feel supported and heard
- Improved trust in administration
- Better student satisfaction with campus services
- Positive feedback from users

Assignment 3: Ideate

Updated Feature Included: Senior & Alumni Interaction Module

Project Topic: PCCOER Student Support Portal

A. Idea Generation

Sr No.	Idea Description	Technique Used	Key Benefit to User
1	Online complaint registration portal	Brainstorming	Easy complaint submission
2	Upload images/videos with complaints	Mind Mapping	Better proof and faster understanding
3	Real-time complaint status tracking	SCAMPER	Improved transparency
4	Suggestion submission module	Brainstorming	Student participation in improvements
5	Mentor/counselor support requests	Mind Mapping	Better student support
6	Ragging complaint section	SCAMPER	Student safety and emergency support
7	Events and hackathons section	Brainstorming	Better awareness of opportunities
8	Campus facilities explorer	Mind Mapping	Easy access to campus information

Sr No.	Idea Description	Technique Used	Key Benefit to User
9	Library book availability system	SCAMPER	Quick access to book information
10	Senior & alumni mentorship interaction system	Brainstorming	Career guidance and networking

B. Mind Map for the Problem

Central Idea:

PCCOER Student Support Portal

Main Branches:

1. Complaint System

- Complaint registration
- Status tracking
- Image/video upload
- Categories

2. Student Support

- Mentor support
- Counselor support
- Ragging complaints

3. Events & Opportunities

- Hackathons
- Workshops
- Club events
- Seminars

4. Campus Facilities

- Gym
- Sports
- Clubs
- Labs
- Canteen

5. Library Module

- Search books
- Check availability
- Rack details

6. Community & Mentorship

- Senior interaction
- Alumni guidance
- Placement experiences
- Mentorship requests

7. Admin Management

- Dashboard
 - Complaint management
 - Event management
 - User management
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C. Decision Matrix

Selected Criteria

- Feasibility
 - User Benefit
 - Innovation
 - Cost Efficiency
 - Ease of Implementation
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D. Define Your Criteria & Weights

Criterion	Weight (1-5)	Description
Feasibility	5	Easy to develop and maintain
User Benefit	5	Improves student experience
Innovation	4	Modern and useful features
Cost Efficiency	4	Affordable implementation
Ease of Implementation	5	Can be completed within project timeline

E. Score Each Idea Against the Criteria

Idea	Idea Description	Feasibility	User Benefit	Innovation	Cost Efficiency	Ease of Implementation	Total Score
1	Complaint portal	5	5	4	5	5	24

Idea	Idea Description	Feasibility	User Benefit	Innovation	Cost Efficiency	Ease of Implementation	Total Score
2	Media upload system	4	5	5	4	4	22
3	Suggestion system	5	4	4	5	5	23
4	Events module	4	5	4	4	4	21
5	Library module	4	4	4	5	4	21

F. Selected Idea

Selected Idea (Brief Description)

A centralized full-stack web portal for PCCOER students and faculty members that provides complaint management, student support services, suggestions, events information, campus facilities details and library book availability through one digital platform.

Why Was This Idea Chosen?

- Provides centralized student support
- Improves transparency and communication
- Reduces manual processes
- Supports real-world campus needs
- Includes modern and useful features
- Feasible within academic project scope
- Enhances student experience and engagement

Why Were Other Ideas Rejected?

- Some ideas were too complex for the project timeline
- Certain features required additional infrastructure and resources
- Some solutions did not align directly with the main objective of student support
- Advanced features such as real-time chat and mobile applications were avoided to maintain project feasibility.

Highlighted Feature: Senior, Mentor & Alumni Interaction Module

Features

This is one of the major highlighted features of the PCCOER Student Support Portal. The module is designed to bridge the communication gap between students, seniors, mentors and alumni by creating a professional guidance and mentorship environment inside the college portal.

- Students can interact with official verified seniors for academic guidance, coding help and placement preparation.
- Alumni can share placement experiences, interview tips and career advice.
- Dedicated internship and placement guidance section.
- Alumni event announcements.
- Mentor and career mentorship support system.
- Seniors can share study materials and tips.

Benefits

The Senior & Alumni Interaction Module makes the portal more student-centric and career-oriented. It helps junior students learn from experienced seniors and alumni, improving placement preparation, technical growth and overall student engagement.

- Improves student networking.
- Helps students with career planning.
- Enhances placement preparation.
- Strengthens alumni-student connection.
- Creates a collaborative college community.